



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
LALITPUR ENGINEERING COLLEGE**

TETRIS GAME WITH DYNAMIC BLOCK MECHANISM

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**A THIRD YEAR PROJECT STUDY REPORT
SUBMITTED TO THE DEPARTMENT OF COMPUTER ENGINEERING
IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR
THE DEGREE OF BACHELOR OF ENGINEERING IN COMPUTER
ENGINEERING**

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**A THIRD YEAR PROJECT STUDY
REPORT ON
TETRIS GAME WITH DYNAMIC BLOCK MECHANISM**

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ABSTRACT

This project presents the design and development of an advanced Tetris game that incorporates a dynamic block mechanism to enhance traditional gameplay. While classic Tetris relies on a fixed set of tetromino shapes and deterministic behavior, this implementation introduces a variety of custom block types with adaptive properties, including shape transformation, autonomous rotation, and physics-inspired movement. These dynamic blocks are procedurally generated and respond to in-game stimuli, creating a more immersive and unpredictable user experience.

The system architecture is modular and object-oriented, allowing for scalable integration of new block behaviors and game features. Core components such as block management, collision detection, rendering, and input handling are decoupled to promote maintainability and extensibility. A fixed timestep game loop ensures consistent performance across platforms, while the rendering pipeline leverages graphics libraries to deliver smooth visual feedback.

In addition to gameplay innovation, the project explores real-time system design principles, including task scheduling, event-driven interactions, and responsive UI handling. The dynamic block mechanism not only increases the strategic complexity of the game but also serves as a platform for experimenting with procedural content generation and interactive simulation. This work demonstrates how classic games can be reimaged through creative extensions and technical rigor, offering both entertainment and insight into modern game development practices.

Keywords: *Collision Detection, Dynamic Blocks, Game Loop, Object-Oriented Design, Procedural Generation, Real-Time Systems, Tetris*

TABLE OF CONTENTS

ACKNOWLEDGMENT	iii
ABSTRACT	iv
TABLE OF CONTENTS	v
LIST OF FIGURES	vii
LIST OF TABLES	viii
LIST OF ABBREVIATIONS	ix
1 INTRODUCTION	1
1.1 Background	1
1.2 Motivation	1
1.3 Problem Statement	1
1.4 Project Objectives	2
1.5 Scope and Application of Project	2
2 REQUIREMENT ANALYSIS	3
2.1 Functional Requirements	3
2.2 Non-Functional Requirements	3
2.3 Hardware Requirements	3
2.4 Software Requirements	3
2.5 Feasibility Study	4
2.5.1 Technical Feasibility	4
2.5.2 Economic Feasibility	4
2.5.3 Operational Feasibility	4
2.5.4 Legal Feasibility	4
3 METHODOLOGY	5
3.1 Overview	5
3.2 Existing vs Proposed System	5
3.3 Software Development Model	6
3.4 Database Table	7
4 System Design	9
4.1 Use Case Diagram	9

4.2	ER Diagram (if applicable)	9
4.3	Data Flow Diagram	10
5	System Testing	13
5.1	Unit Testing	13
5.2	Integration Testing	13
5.3	System Testing	14
5.4	Whitebox Testing	15
5.5	Blackbox Testing	16
6	Discussion	34
7	Conclusion and Future Enhancements	35
7.1	Conclusion	35
7.2	Limitations	35
7.3	Future Enhancements	35
	REFERENCES	36
	REFERENCES	36

LIST OF FIGURES

Figure 4.1	Use Case Diagram showing Player interaction and Platform dependency	9
Figure 4.2	Entity-Relationship Diagram for User and Score Management	10
Figure 4.3	Level 0 DFD – Context Diagram	11
Figure 4.4	Level 1 DFD – System Overview	11
Figure 4.5	Level 2 DFD – Internal Module Interactions	12
Figure 5.1	Tetris Game	32
Figure 5.2	Tetris Game	33

LIST OF TABLES

Table 3.1	Sample Database Table for Score Tracking	8
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LIST OF ABBREVIATIONS

API	Application Programming Interface
CPU	Central Processing Unit
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
FPS	Frames Per Second
GUI	Graphical User Interface
IDE	Integrated Development Environment
OOP	Object-Oriented Programming
RAM	Random Access Memory
UML	Unified Modeling Language
UI	User Interface

1 INTRODUCTION

1.1 Background

Tetris is a globally recognized puzzle game that has captivated players for decades with its simple yet addictive mechanics. Traditionally, the game involves manipulating falling tetrominoes to form complete horizontal lines within a fixed grid. Once a line is completed, it disappears, and the player earns points. Despite its minimalist design, Tetris has been widely studied in fields such as cognitive science, artificial intelligence, and game theory due to its elegant balance of strategy and reflex.

Modern game development, however, increasingly emphasizes dynamic and adaptive gameplay to enhance user engagement. This project builds upon the foundational mechanics of Tetris by introducing a dynamic block mechanism, where blocks exhibit behaviors beyond static movement and rotation. These enhancements aim to create a more immersive and technically challenging experience while exploring real-time system design and procedural content generation.

1.2 Motivation

The motivation behind this project stems from the desire to reimagine a classic game through the lens of modern programming paradigms and interactive design. By integrating dynamic block behaviors—such as shape transformation, autonomous rotation, and physics-based movement—the game becomes a platform for experimenting with real-time mechanics, modular architecture, and user-centric innovation. This project also serves as a practical application of object-oriented programming, graphics rendering, and system-level design principles, making it an ideal candidate for academic exploration and portfolio development.

1.3 Problem Statement

Traditional Tetris games are limited by their fixed set of block shapes and predictable behavior, which can lead to repetitive gameplay and reduced long-term engagement. There is a need for a more flexible and adaptive system that allows for procedural generation of blocks, dynamic interactions, and scalable architecture. The challenge lies in maintaining the core appeal of Tetris while introducing complexity in a controlled and user-friendly manner.

1.4 Project Objectives

The main objectives of this project are:

- To design and implement a Tetris game with dynamic block mechanisms that extend traditional gameplay.
- To develop a modular and object-oriented architecture for managing game components.
- To integrate real-time system features such as fixed timestep loops, event-driven interactions, and responsive rendering.
- To explore procedural content generation for block behavior and game difficulty.
- To create an engaging user experience through visual feedback and adaptive gameplay.

1.5 Scope and Application of Project

This project is primarily focused on the development of a desktop-based Tetris game using graphics libraries such as OpenGL or WebGL. The scope includes block design, game loop implementation, collision detection, and user interface integration. While the core mechanics are inspired by classic Tetris, the dynamic block system introduces novel gameplay elements suitable for both casual players and technical demonstrations.

Applications of this project extend to educational tools for teaching programming and system design, as well as prototypes for interactive simulations and game development frameworks. The modular nature of the system allows for future expansion into mobile platforms, multiplayer modes, or AI-driven gameplay enhancements.

2 REQUIREMENT ANALYSIS

2.1 Functional Requirements

- The system shall allow users to play a Tetris game with dynamically generated blocks.
- The game shall support real-time block movement, rotation, and interaction.
- The system shall detect collisions and clear completed lines.
- The game shall progressively increase difficulty based on player performance.
- The system shall provide visual feedback for block behavior and game events.
- The user shall be able to pause, resume, and restart the game.

2.2 Non-Functional Requirements

- The game shall maintain a consistent frame rate across platforms.
- The system shall be modular and extensible for future enhancements.
- The interface shall be responsive and intuitive for users.
- The game shall be compatible with major desktop operating systems.
- The system shall be developed using open-source tools and libraries.

2.3 Hardware Requirements

- Standard desktop or laptop with keyboard input.
- Optional: Raspberry Pi for embedded deployment or experimentation.
- Optional: External display or controller for extended gameplay.

2.4 Software Requirements

- Programming Language: C++, Python, or JavaScript
- Graphics Library: OpenGL, WebGL, or Pygame
- Development Environment: VS Code, Code::Blocks, or browser-based IDE

- Version Control: Git and GitHub
- Optional: SDL or GLFW for input and window management

2.5 Feasibility Study

2.5.1 Technical Feasibility

The technologies required for this project—such as OpenGL for rendering, object-oriented programming for modular design, and fixed timestep loops for real-time control—are well-established and widely supported. The system architecture is designed to be platform-independent, allowing development and testing on standard desktop environments. Procedural generation and dynamic block behavior can be implemented using existing libraries and algorithms, making the technical execution highly viable.

2.5.2 Economic Feasibility

The project leverages open-source tools and frameworks, minimizing development costs. No specialized hardware is required beyond a standard computer, and optional components like Raspberry Pi are low-cost and accessible. The economic feasibility is strong, especially for academic or prototype-level deployment.

2.5.3 Operational Feasibility

The game is designed to be user-friendly, with intuitive controls and visual feedback. Its modular architecture allows for easy updates and maintenance. The system can be operated by users with basic gaming experience and does not require technical expertise to play. From a developer's perspective, the codebase is structured for readability and scalability.

2.5.4 Legal Feasibility

The project uses open-source libraries and adheres to standard licensing practices. No proprietary assets or restricted technologies are involved. As long as proper attribution is maintained for third-party tools, the project complies with legal and ethical standards for software development.

3 METHODOLOGY

3.1 Overview

The methodology adopted for this project follows a modular and iterative approach to game development. The system is divided into distinct components, each responsible for a specific function within the game loop. The order of execution is as follows:

1. **Initialization:** Load game assets, initialize grid, and configure system parameters.
2. **Input Handling:** Capture user inputs (e.g., movement, rotation, pause) via keyboard or controller.
3. **Block Generation:** Spawn a new block with dynamic properties such as shape, rotation behavior, or transformation logic.
4. **Game Loop Execution:**
 - Apply gravity and movement rules.
 - Check for collisions and boundary constraints.
 - Update block position and state.
 - Detect line completion and clear rows.
 - Trigger dynamic behaviors (e.g., shape shift, split, merge).
5. **Rendering:** Refresh the visual display using graphics libraries (e.g., OpenGL/WebGL).
6. **Score and Level Update:** Adjust score and difficulty based on gameplay events.
7. **Game Over Check:** Determine if blocks have reached the top of the grid.

Each component is encapsulated within its own module, allowing for independent testing and future scalability.

3.2 Existing vs Proposed System

Existing System:

- Fixed set of tetromino shapes with predefined rotation states

- Static block behavior with no transformation or interaction
- Predictable gameplay with limited variation
- Monolithic architecture with tightly coupled components
- Basic collision detection and line clearing logic

Proposed System:

- Procedurally generated blocks with dynamic shapes and behaviors
- Autonomous rotation, transformation, and physics-based movement
- Adaptive gameplay with unpredictable block interactions
- Modular architecture for scalable and maintainable development
- Advanced collision handling and behavior-triggered responses

3.3 Software Development Model

The software development model adopted for this project is the **Iterative Development Model**, which emphasizes incremental progress through repeated cycles of design, implementation, testing, and refinement. This approach is particularly well-suited for game development projects where gameplay mechanics, user experience, and system performance evolve over time.

Unlike the traditional Waterfall model, which requires all requirements to be finalized before development begins, the iterative model allows for continuous feedback and adaptation. Each iteration focuses on a specific set of features or improvements, enabling the developer to test and validate components in isolation before integrating them into the full system.

Justification for Iterative Model

- **Flexibility:** Dynamic block behaviors and procedural generation require experimentation and tuning, which is best supported by an adaptive development cycle.

- **Modularity:** The game is built using a modular architecture, allowing individual components (e.g., block manager, renderer, input handler) to be developed and tested independently.
- **Rapid Prototyping:** New gameplay features can be quickly implemented and evaluated without disrupting the entire system.
- **User Feedback Integration:** Visual and interactive elements benefit from iterative refinement based on user testing and performance analysis.
- **Risk Reduction:** Early detection of design flaws or performance bottlenecks helps avoid costly rework later in the development cycle.

Development Phases

The project progresses through the following iterative phases:

1. **Requirement Analysis:** Identify core gameplay features, dynamic block behaviors, and system constraints.
2. **Design:** Define system architecture, data structures, and interaction flow.
3. **Implementation:** Develop modules such as block generation, collision detection, rendering, and input handling.
4. **Testing:** Validate functionality, performance, and user experience through unit and integration testing.
5. **Evaluation:** Analyze gameplay dynamics, gather feedback, and refine mechanics.
6. **Iteration:** Repeat the cycle to introduce new features or improve existing ones.

This model ensures that the final product is both technically robust and engaging, while allowing for creative exploration throughout the development process.

3.4 Database Table

Although Tetris is primarily a logic-based game, a lightweight database can be used to store user scores, game settings, and session history. Below is a sample table structure:

Field Name	Data Type	Description	Constraints
UserID	INT	Unique identifier for player	Primary Key
Username	VARCHAR(50)	Player name	Not Null
Score	INT	Highest score achieved	Default = 0
LevelReached	INT	Last level completed	Default = 1
PlayDate	DATE	Date of last session	-

Table 3.1: Sample Database Table for Score Tracking

4 System Design

4.1 Use Case Diagram

A Use Case Diagram is a type of UML (Unified Modeling Language) diagram that visually represents the interactions between actors (users or external systems) and a system to achieve specific goals.

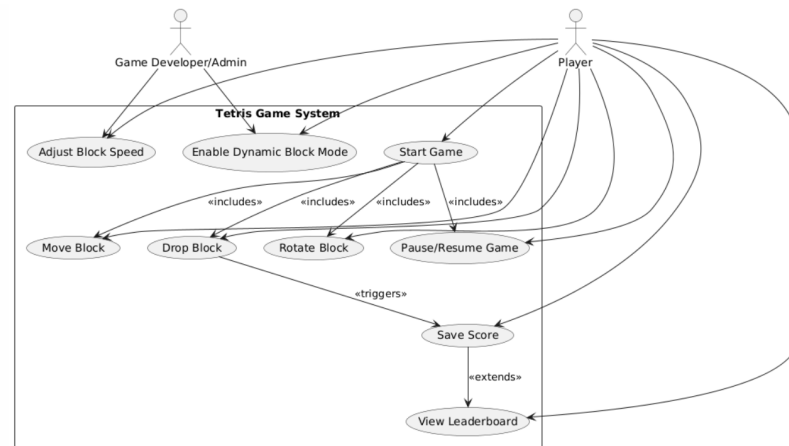


Figure 4.1: Use Case Diagram showing Player interaction and Platform dependency

4.2 ER Diagram (if applicable)

If a database is implemented to store user data and scores, the following Entity-Relationship (ER) diagram represents the logical structure of the system. It illustrates the key entities, their attributes, and the relationships between them—ensuring efficient data management and retrieval.

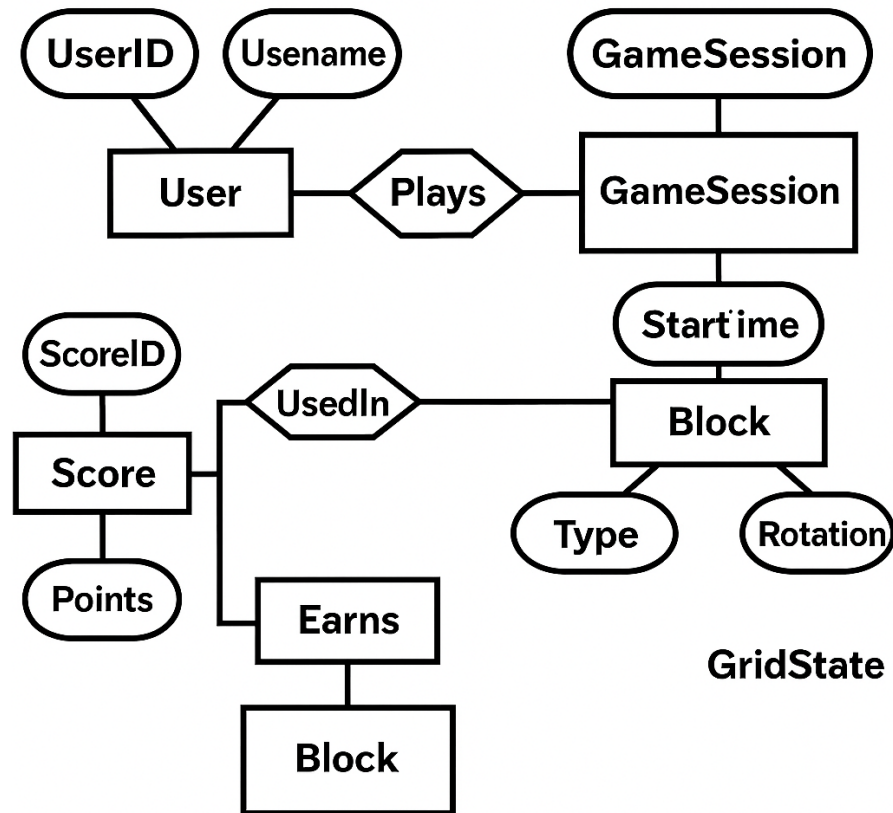


Figure 4.2: Entity-Relationship Diagram for User and Score Management

Entities:

- **User:** Stores player information such as UserID, Username, and PlayDate.
- **GameSession:** Records score, level reached, and session duration.

4.3 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the system during gameplay. It highlights the interaction between user inputs, game logic, and visual output. The diagrams are organized into three levels to represent increasing detail and modularity.

Level 0 – Context Diagram

This high-level diagram represents the entire game system as a single process. It shows interactions with external entities such as the player and optional data storage.

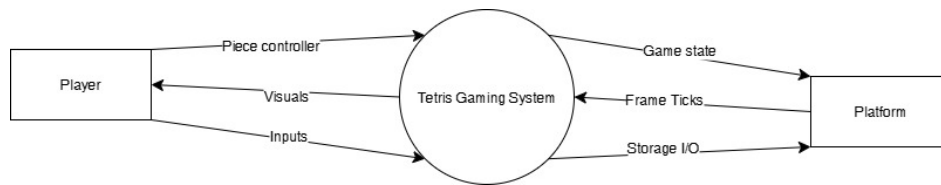


Figure 4.3: Level 0 DFD – Context Diagram

Level 1 – System Overview

This diagram decomposes the main system into core functional units such as Input Handler, Game Engine, and Display Manager. It shows how data flows between these subsystems.

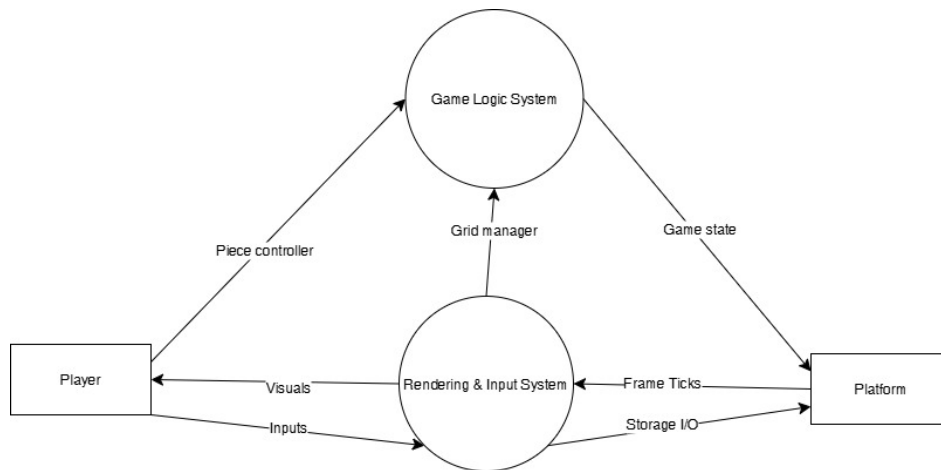


Figure 4.4: Level 1 DFD – System Overview

Level 2 – Internal Module Interactions

This detailed diagram breaks down individual modules like Block Manager, Collision Detector, Score Tracker, and Renderer. It captures internal data transformations and control flow.

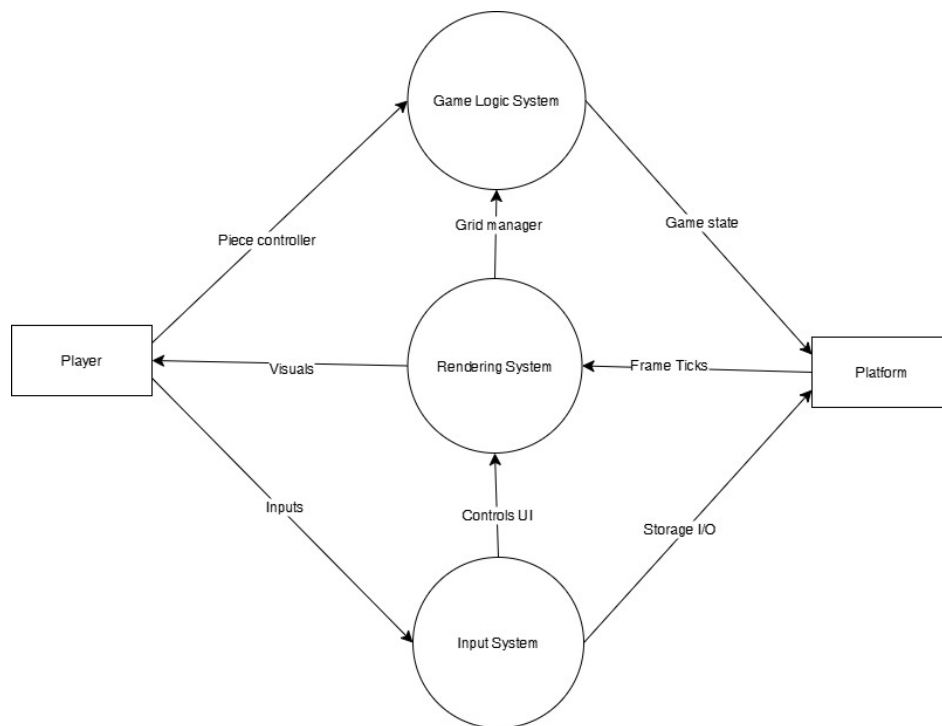


Figure 4.5: Level 2 DFD – Internal Module Interactions

DFD Description:

- **User Input:** Captured and processed by the Input Handler.
- **Block Manager:** Generates and updates block states.
- **Collision Detector:** Validates movements and triggers line clearing.
- **Renderer:** Displays the updated game state.
- **Score Manager:** Updates progress and stores data if applicable.

5 System Testing

System testing is a critical phase in the software development lifecycle where the complete and integrated system is evaluated to ensure it meets the specified requirements. It encompasses various testing strategies to validate both functional and non-functional aspects of the software.

5.1 Unit Testing

Unit testing involves testing individual components or modules of the software in isolation. Each unit is verified to ensure it performs as expected.

- **Objective:** Validate the correctness of individual functions, methods, or classes.
- **Tools Used:** Custom test scripts, assertions, debugging tools.
- **Example:** Testing the block rotation logic in the Tetris game to ensure it adheres to the defined transformation rules.

Source Code

```
// Test for block rotation
Block block = new Block('T');
block.rotate();
assert(block.getOrientation() == expectedOrientation);
```

Expected Output

Test Passed: Block rotated to expected orientation.

5.2 Integration Testing

Integration testing focuses on verifying the interactions between integrated modules. It ensures that data flows correctly and modules cooperate as intended.

- **Objective:** Detect interface defects and ensure smooth data exchange between modules.
- **Approach:** Incremental integration using stubs and drivers.

- **Example:** Testing the interaction between the game logic module and the rendering engine in the Tetris system.

Source Code

```
// Stub for rendering engine
class RenderStub {
    void draw(Block block) {
        System.out.println("Drawing block: " + block.getType());
    }
}

// Integration test
GameEngine engine = new GameEngine(new RenderStub());
engine.spawnBlock('L');
```

Expected Output

```
Drawing block: L
Block spawned successfully.
```

5.3 System Testing

System testing validates the entire software system as a whole. It checks for compliance with functional and non-functional requirements.

- **Objective:** Ensure the complete system behaves as expected under various conditions.
- **Scope:** Includes performance, usability, reliability, and security testing.
- **Example:** Running the Tetris game on different platforms to verify consistent behavior and responsiveness.

Source Code

```
// Launch game on multiple platforms
```

```
SystemTest.runOnPlatform("Windows");  
SystemTest.runOnPlatform("Linux");  
SystemTest.runOnPlatform("Android");
```

Expected Output

```
Windows: Game launched successfully.  
Linux: Game launched successfully.  
Android: Game launched successfully.  
All platforms passed system test.
```

5.4 Whitebox Testing

Whitebox testing, also known as structural testing, involves examining the internal logic and structure of the code.

- **Objective:** Verify internal operations such as loops, branches, and data flow.
- **Techniques:** Statement coverage, branch coverage, path coverage.
- **Example:** Analyzing the scoring algorithm in the Tetris game to ensure all conditional paths are tested.

Source Code

```
// Score calculation logic  
int calculateScore(int linesCleared) {  
    if (linesCleared == 1) return 100;  
    else if (linesCleared == 2) return 300;  
    else if (linesCleared == 3) return 500;  
    else return 800;  
}
```

Expected Output

```
Input: 2 lines cleared  
Output: 300 points  
Path coverage: All branches tested
```


5.5 Blackbox Testing

Blackbox testing evaluates the system based on inputs and expected outputs without knowledge of internal code structure.

- **Objective:** Validate functional behavior against requirements.
- **Techniques:** Equivalence partitioning, boundary value analysis, decision table testing.
- **Example:** Testing user input for block movement and verifying correct game response without inspecting the code.

Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Tetris Game</title>
  <style>
    body {
      margin: 0;
      padding: 20px;
      background: linear-gradient(135deg, #1e3c72, #2a5298);
      font-family: 'Courier New', monospace;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      color: white;
    }

    .game-container {
```

```

        display: flex;
        gap: 30px;
        align-items: flex-start;
    }

    .game-board {
        border: 4px solid #fff;
        border-radius: 10px;
        box-shadow: 0 0 30px rgba(255, 255, 255, 0.3);
        background: #000;
    }

    .info-panel {
        display: flex;
        flex-direction: column;
        gap: 20px;
        min-width: 200px;
    }

    .score-box, .next-box, .controls-box {
        background: rgba(255, 255, 255, 0.1);
        border-radius: 10px;
        padding: 20px;
        backdrop-filter: blur(10px);
        border: 1px solid rgba(255, 255, 255, 0.2);
    }

    .score-box h3, .next-box h3, .controls-box h3 {
        margin: 0 0 15px 0;
        color: #fff;
        text-shadow: 0 0 10px rgba(255, 255, 255, 0.5);
    }

```

```
.next-preview {  
    width: 80px;  
    height: 80px;  
    background: #000;  
    border: 2px solid #333;  
    border-radius: 5px;  
    margin: 10px 0;  
}
```

```
.controls-box p {  
    margin: 5px 0;  
    font-size: 12px;  
    opacity: 0.8;  
}
```

```
.game-over {  
    position: absolute;  
    top: 50%;  
    left: 50%;  
    transform: translate(-50%, -50%);  
    background: rgba(0, 0, 0, 0.9);  
    color: white;  
    padding: 30px;  
    border-radius: 15px;  
    text-align: center;  
    border: 2px solid #fff;  
    display: none;  
}
```

```
button {  
    background: linear-gradient(45deg, #ff6b6b, #ee5a52);
```

```

        border: none;
        color: white;
        padding: 10px 20px;
        border-radius: 25px;
        cursor: pointer;
        font-family: inherit;
        margin-top: 15px;
        transition: transform 0.2s;
    }

    button:hover {
        transform: scale(1.05);
    }

    .pulse {
        animation: pulse 0.2s;
    }

    @keyframes pulse {
        0% { transform: scale(1); }
        50% { transform: scale(1.05); }
        100% { transform: scale(1); }
    }
</style>
</head>
<body>
    <div class="game-container">
        <canvas id="gameBoard" class="game-board" width="300" height="600"></canvas>

        <div class="info-panel">
            <div class="score-box">
                <h3>Score</h3>
            </div>
        </div>
    </div>

```

```

        <div id="score">0</div>
        <h3>Level</h3>
        <div id="level">1</div>
        <h3>Lines</h3>
        <div id="lines">0</div>
    </div>

    <div class="next-box">
        <h3>Next</h3>
        <canvas id="nextPreview" class="next-preview" width="80" height="80"></canvas>
    </div>

    <div class="controls-box">
        <h3>Controls</h3>
        <p>← → : Move</p>
        <p>↓ : Soft Drop</p>
        <p>↑ : Rotate</p>
        <p>Space : Hard Drop</p>
        <p>P : Pause</p>
    </div>
</div>

<div id="gameOver" class="game-over">
    <h2>Game Over!</h2>
    <p>Final Score: <span id="finalScore">0</span></p>
    <button onclick="startNewGame()">Play Again</button>
</div>

<script>
    const canvas = document.getElementById('gameBoard');
    const ctx = canvas.getContext('2d');
```

```

const nextCanvas = document.getElementById('nextPreview');
const nextCtx = nextCanvas.getContext('2d');

const BOARD_WIDTH = 10;
const BOARD_HEIGHT = 20;
const BLOCK_SIZE = 30;

let board = [];
let currentPiece = null;
let nextPiece = null;
let score = 0;
let level = 1;
let lines = 0;
let dropTime = 0;
let dropInterval = 1000;
let gameRunning = false;
let paused = false;

// Tetris pieces
const pieces = [
  {
    shape: [
      [1, 1, 1, 1]
    ],
    color: '#00f5ff'
  },
  {
    shape: [
      [1, 1],
      [1, 1]
    ],
    color: '#ffff00'
  }
]

```

```

},
{
    shape: [
        [0, 1, 0],
        [1, 1, 1]
    ],
    color: '#a020f0'
},
{
    shape: [
        [1, 0, 0],
        [1, 1, 1]
    ],
    color: '#ff8c00'
},
{
    shape: [
        [0, 0, 1],
        [1, 1, 1]
    ],
    color: '#0000ff'
},
{
    shape: [
        [1, 1, 0],
        [0, 1, 1]
    ],
    color: '#00ff00'
},
{
    shape: [
        [0, 1, 1],

```

```

        [1, 1, 0]
    ],
    color: '#ff0000'
}
];

function initBoard() {
    board = [];
    for (let y = 0; y < BOARD_HEIGHT; y++) {
        board[y] = [];
        for (let x = 0; x < BOARD_WIDTH; x++) {
            board[y][x] = 0;
        }
    }
}

function createPiece() {
    const pieceIndex = Math.floor(Math.random() * pieces.length);
    const piece = JSON.parse(JSON.stringify(pieces[pieceIndex]));
    piece.x = Math.floor(BOARD_WIDTH / 2) - Math.floor(piece.shape[0].length / 2);
    piece.y = 0;
    return piece;
}

function drawBlock(ctx, x, y, color) {
    const gradient = ctx.createLinearGradient(x * BLOCK_SIZE, y * BLOCK_SIZE,
                                              (x + 1) * BLOCK_SIZE, (y + 1) * BLOCK_SIZE);
    gradient.addColorStop(0, color);
    gradient.addColorStop(1, darkenColor(color, 0.3));

    ctx.fillStyle = gradient;
    ctx.fillRect(x * BLOCK_SIZE, y * BLOCK_SIZE, BLOCK_SIZE, BLOCK_SIZE);
}

```



```

    // Add shine effect
    ctx.fillStyle = 'rgba(255, 255, 255, 0.3)';
    ctx.fillRect(x * BLOCK_SIZE + 2, y * BLOCK_SIZE + 2, BLOCK_SIZE - 1, BLOCK_SIZE - 1);

    // Border
    ctx.strokeStyle = darkenColor(color, 0.5);
    ctx.lineWidth = 2;
    ctx.strokeRect(x * BLOCK_SIZE, y * BLOCK_SIZE, BLOCK_SIZE, BLOCK_SIZE);
  }

function darkenColor(color, amount) {
  const num = parseInt(color.replace("#", ""), 16);
  const amt = Math.round(255 * amount);
  const R = Math.max(0, (num >> 16) - amt);
  const G = Math.max(0, (num >> 8 & 0x00FF) - amt);
  const B = Math.max(0, (num & 0x0000FF) - amt);
  return "#" + ((1 << 24) + (R << 16) + (G << 8) + B).toString(16).slice(1);
}

function drawBoard() {
  ctx.fillStyle = '#000';
  ctx.fillRect(0, 0, canvas.width, canvas.height);

  // Draw placed blocks
  for (let y = 0; y < BOARD_HEIGHT; y++) {
    for (let x = 0; x < BOARD_WIDTH; x++) {
      if (board[y][x]) {
        drawBlock(ctx, x, y, board[y][x]);
      }
    }
  }
}

```

```

// Draw current piece
if (currentPiece) {
    for (let y = 0; y < currentPiece.shape.length; y++) {
        for (let x = 0; x < currentPiece.shape[y].length; x++) {
            if (currentPiece.shape[y][x]) {
                drawBlock(ctx, currentPiece.x + x, currentPiece.y + y,
                           currentPiece.shape[y][x]);
            }
        }
    }
}
}

```

```

function drawNextPiece() {
    nextCtx.fillStyle = '#000';
    nextCtx.fillRect(0, 0, nextCanvas.width, nextCanvas.height);

    if (nextPiece) {
        const blockSize = 15;
        const offsetX = (nextCanvas.width - nextPiece.shape[0].length *
                           blockSize) / 2;
        const offsetY = (nextCanvas.height - nextPiece.shape.length *
                           blockSize) / 2;

        for (let y = 0; y < nextPiece.shape.length; y++) {
            for (let x = 0; x < nextPiece.shape[y].length; x++) {
                if (nextPiece.shape[y][x]) {
                    nextCtx.fillStyle = nextPiece.color;
                    nextCtx.fillRect(offsetX + x * blockSize, offsetY + y *
                                      blockSize,
                                      blockSize, blockSize);
                    nextCtx.strokeStyle = '#333';
                    nextCtx.strokeRect(offsetX + x * blockSize, offsetY + y *
                                        blockSize,
                                        blockSize, blockSize);
                }
            }
        }
    }
}

```

```

    }
  }
}

function canMove(piece, dx, dy) {
  for (let y = 0; y < piece.shape.length; y++) {
    for (let x = 0; x < piece.shape[y].length; x++) {
      if (piece.shape[y][x]) {
        const newX = piece.x + x + dx;
        const newY = piece.y + y + dy;

        if (newX < 0 || newX >= BOARD_WIDTH || newY >= BOARD_HEIGHT) {
          return false;
        }

        if (newY >= 0 && board[newY][newX]) {
          return false;
        }
      }
    }
  }
  return true;
}

```

```

function placePiece() {
  for (let y = 0; y < currentPiece.shape.length; y++) {
    for (let x = 0; x < currentPiece.shape[y].length; x++) {
      if (currentPiece.shape[y][x]) {
        const boardY = currentPiece.y + y;
        const boardX = currentPiece.x + x;

```

```

        if (boardY < 0) {
            gameOver();
            return;
        }

        board[boardY][boardX] = currentPiece.color;
    }
}

clearLines();
currentPiece = nextPiece;
nextPiece = createPiece();
}

function clearLines() {
    let linesCleared = 0;

    for (let y = BOARD_HEIGHT - 1; y >= 0; y--) {
        let fullLine = true;
        for (let x = 0; x < BOARD_WIDTH; x++) {
            if (!board[y][x]) {
                fullLine = false;
                break;
            }
        }

        if (fullLine) {
            // Add flash effect
            canvas.classList.add('pulse');
            setTimeout(() => canvas.classList.remove('pulse'), 200);
        }
    }
}

```

```

        board.splice(y, 1);
        board.unshift(new Array(BOARD_WIDTH).fill(0));
        linesCleared++;
        y++; // Check same line again
    }
}

if (linesCleared > 0) {
    lines += linesCleared;
    score += linesCleared * 100 * level;
    level = Math.floor(lines / 10) + 1;
    dropInterval = Math.max(100, 1000 - (level - 1) * 50);
    updateDisplay();
}
}

function rotatePiece() {
    const rotated = [];
    const rows = currentPiece.shape.length;
    const cols = currentPiece.shape[0].length;

    for (let x = 0; x < cols; x++) {
        rotated[x] = [];
        for (let y = 0; y < rows; y++) {
            rotated[x][y] = currentPiece.shape[rows - 1 - y][x];
        }
    }

    const originalShape = currentPiece.shape;
    currentPiece.shape = rotated;

    if (!canMove(currentPiece, 0, 0)) {

```

```

        currentPiece.shape = originalShape;
    }
}

function hardDrop() {
    while (canMove(currentPiece, 0, 1)) {
        currentPiece.y++;
        score += 2;
    }
    placePiece();
    updateDisplay();
}

function updateDisplay() {
    document.getElementById('score').textContent = score;
    document.getElementById('level').textContent = level;
    document.getElementById('lines').textContent = lines;
}

function gameOver() {
    gameRunning = false;
    document.getElementById('finalScore').textContent = score;
    document.getElementById('gameOver').style.display = 'block';
}

function startNewGame() {
    initBoard();
    currentPiece = createPiece();
    nextPiece = createPiece();
    score = 0;
    level = 1;
    lines = 0;
}

```

```

    dropTime = 0;
    dropInterval = 1000;
    gameRunning = true;
    paused = false;
    document.getElementById('gameOver').style.display = 'none';
    updateDisplay();
    gameLoop();
}

```

```

function gameLoop() {
    if (!gameRunning || paused) return;

    const now = Date.now();
    if (now - dropTime > dropInterval) {
        if (canMove(currentPiece, 0, 1)) {
            currentPiece.y++;
        } else {
            placePiece();
        }
        dropTime = now;
    }
}

```

```

drawBoard();
drawNextPiece();

requestAnimationFrame(gameLoop);
}

```

```

// Controls
document.addEventListener('keydown', (e) => {
    if (!gameRunning || paused) {
        if (e.key === 'p' || e.key === 'P') {

```

```

        paused = !paused;
        if (!paused) gameLoop();
    }
    return;
}

switch (e.key) {
    case 'ArrowLeft':
        if (canMove(currentPiece, -1, 0)) {
            currentPiece.x--;
        }
        break;
    case 'ArrowRight':
        if (canMove(currentPiece, 1, 0)) {
            currentPiece.x++;
        }
        break;
    case 'ArrowDown':
        if (canMove(currentPiece, 0, 1)) {
            currentPiece.y++;
            score += 1;
            updateDisplay();
        }
        break;
    case 'ArrowUp':
        rotatePiece();
        break;
    case ' ':
        e.preventDefault();
        hardDrop();
        break;
    case 'p':

```



```

        case 'P':
            paused = !paused;
            if (!paused) gameLoop();
            break;
    }
});

// Start the game
startNewGame();
</script>
</body>
</html>

```

Output

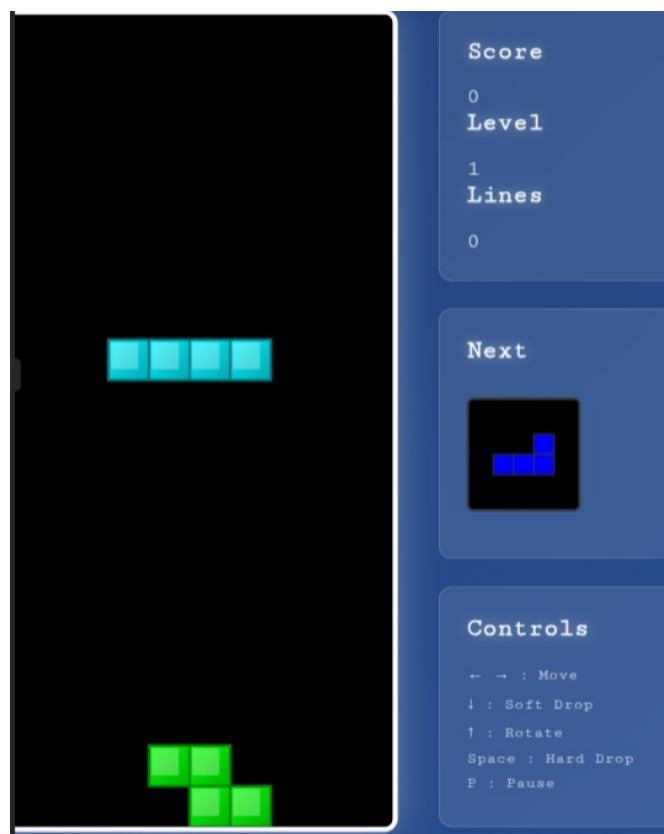


Figure 5.1: Tetris Game

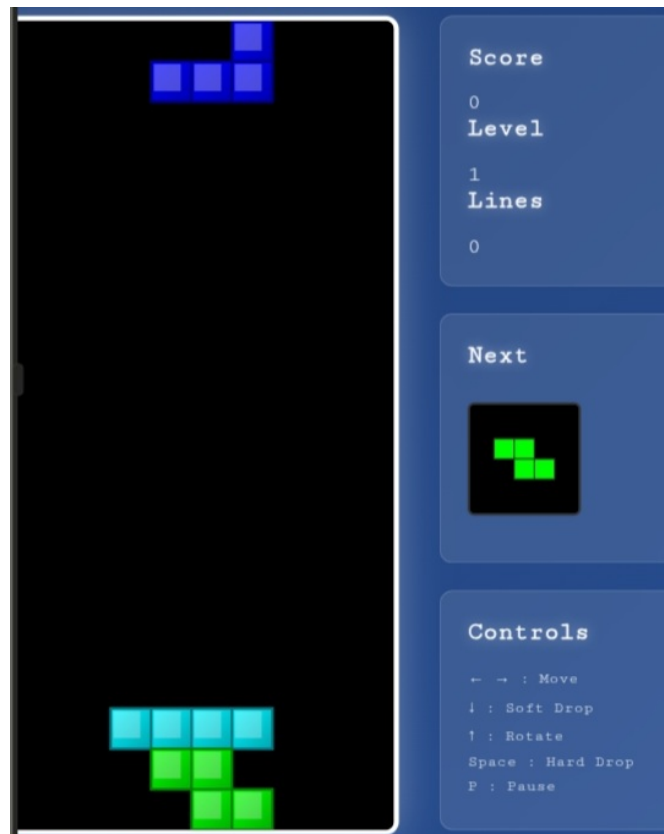


Figure 5.2: Tetris Game

6 Discussion

The development of the modular Tetris game project reflects a comprehensive approach to software design, emphasizing architectural clarity, functional robustness, and maintainability. By adopting a modular structure, the system was divided into distinct components such as block generation, collision detection, rendering, and scoring—each designed to operate independently yet cohesively. This separation of concerns greatly facilitated debugging, iterative development, and future scalability. The integration of system diagrams, including Entity-Relationship (ER) and Data Flow Diagrams (DFDs), provided valuable insights into the logical structure and data movement within the system, enhancing both documentation and stakeholder communication. Rigorous testing methodologies were applied throughout the development lifecycle, including unit testing for individual modules, integration testing for subsystem interactions, and system testing to validate overall behavior. Whitebox testing helped uncover edge cases in the scoring and rotation logic, while blackbox testing ensured that user inputs and game responses aligned with expected outcomes. Several challenges were encountered, particularly in synchronizing real-time rendering with game logic, implementing precise collision detection across varied block shapes and orientations, and ensuring consistent performance across multiple platforms. These obstacles were addressed through careful design and iterative refinement. Looking forward, the system offers a solid foundation for enhancements such as multiplayer functionality, cloud-based score tracking, AI-driven gameplay, and improved accessibility features. Such additions would not only expand the game's capabilities but also increase its appeal to a broader user base. Overall, the project stands as a technically sound and well-documented implementation of a classic game, demonstrating thoughtful design, effective testing, and clear potential for future innovation.

7 Conclusion and Future Enhancements

7.1 Conclusion

The primary objective of this project was to design and implement a modular, technically robust version of the classic Tetris game, with a focus on clarity, scalability, and documentation. Through systematic design methodologies—including use case modeling, ER and DFD diagrams, and layered testing strategies—the project successfully delivered a functional and well-structured application. Each module, from block generation to collision detection and scoring, was developed with attention to modularity and maintainability. The outcomes demonstrate a reliable and responsive gameplay experience, supported by thorough testing and clear architectural planning.

7.2 Limitations

While the project achieved its core goals, several limitations remain. The current implementation does not support multiplayer functionality or cloud-based score persistence. Platform-specific optimizations, such as mobile responsiveness and touch controls, were not fully explored. Additionally, advanced features like AI opponents, accessibility customization, and real-time analytics were beyond the scope of this phase.

7.3 Future Enhancements

To extend the system's capabilities and usability, the following enhancements are proposed:

- **Integration with mobile apps:** Develop platform-specific interfaces and controls for Android and iOS devices.
- **Real-time streaming data support:** Enable live score tracking, analytics, and gameplay telemetry for competitive or educational use.
- **Deployment on edge devices:** Optimize the game for low-latency execution on edge platforms such as Raspberry Pi or embedded systems.

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